

Answer ALL questions.

Write your answers in the spaces provided.

- 1** An investigation compared the protein content of some foods.

Each food was crushed in distilled water to produce a suspension. This was filtered and the liquid filtrate was tested for protein.

- (a) (i) Describe how you could test for protein in this filtrate.

(2)

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- (ii) Describe how the filtrates should have been prepared to allow a valid comparison of the protein content of these foods.

(3)

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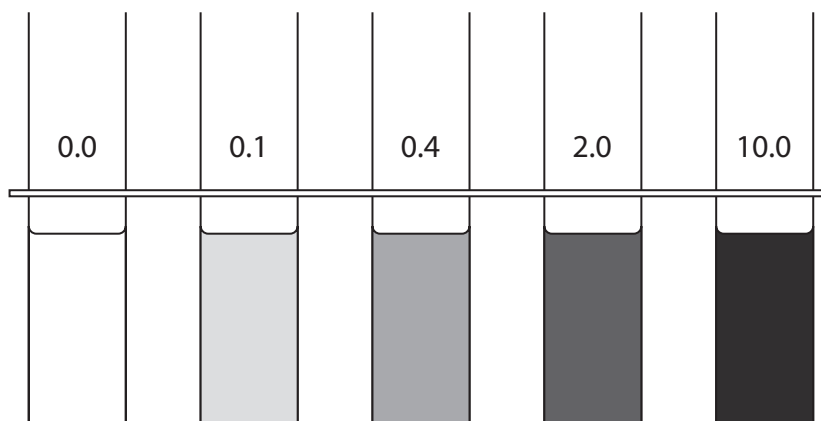
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- (iii) Protein solutions of concentrations from 0.0 to 10.0 mg cm^{-3} were tested using a semi-quantitative method.
The results of the test are shown in the diagram.



Describe the difference between a semi-quantitative and a quantitative test.

(2)

- (iv) Explain how the results shown in the diagram can be used to compare the protein concentrations of food extracts.

(2)



- (b) Another method used to determine the protein content of food is to measure the total nitrogen content. The total nitrogen content is multiplied by a conversion factor to give protein content.

The table shows data for milk and soya beans.

Food	Total nitrogen content / mg	Conversion factor	Protein content / mg
milk	505	6.38	
soya beans	3 600	5.71	20 556

- (i) Use the information in the table to calculate the ratio of the protein content of soya beans to that of milk.

Give your answer to one decimal place.

(2)

Answer

- (ii) Some of the total nitrogen content of a food is not due to protein.

Give **one** other type of organic molecule that contains nitrogen.

(1)

(Total for Question 1 = 12 marks)

